

Project Initialization and Planning Phase

Date	21 September 2026
Team ID	MCA
Project Title	Smartphone Addiction Analytics Dashboard
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Analyze demographic, behaviour and lifestyle attributes from smartphone usage data to uncover patterns, identify high-risk user segments, and determine the strongest predictors of smartphone addiction.
Scope	Build an interactive analytical dashboard covering age, gender, daily/weekend screen time, social media & gaming hours, notifications, app opens, sleep, stress, academic/work impact and addiction level. Focus is on descriptive and diagnostic analytics for product, research and well-being stakeholders. Predictive modelling and real-time mobile integration are out of scope.
Problem Statement	
Description	Smartphone usage behaviour is driven by many interacting factors (screen time, notifications, sleep, stress, academic

	impact). Without a consolidated view, product teams, researchers and digital well-being programmes overlook high-risk segments, misjudge which behaviour drive addiction, and miss opportunities for early intervention.
Impact	Solving this enables stakeholders to reduce the share of users in the Severe addiction band, prioritise feature investment (usage limits, mindful notifications), design targeted well-being nudges, flag early warning signs, and provide evidence-based digital well-being thresholds.
Proposed Solution	
Approach	Load and clean the Smartphone Usage dataset (7,500 records). Perform exploratory data analysis and correlation analysis. Build interactive visualisation and filters (age, gender, stress, academic impact). Deliver persona-specific views for product managers, behaviour analysts and digital-health researchers.
Key Features	1. Overview KPIs (Severe addiction %, average screen time, high-stress users). 2. Segment comparison by age/gender/stress/academic impact. 3. behaviour insight charts (gaming, sleep, work/study vs addiction). 4. Predictor ranking (app opens, notifications, weekend screen time). 5. Interactive filters and drill-down for high-risk segment identification.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for data processing and dashboard serving	4–8 core CPU (no GPU required for this analytics scope)

Memory	RAM for data loading and interactive queries	16 GB RAM
Storage	Disk space for dataset, processed files and logs	50 GB SSD
Software		
Frameworks	Dashboard & analytics frameworks	Streamlit / Dash / Power BI or Tableau
Libraries	Data analysis & visualisation libraries	pandas, numpy, plotly, seaborn, scikit-learn
Development Environment	IDE and version control	VS Code / Jupyter Lab, Git + GitHub
Data		
Data	Source, size and format of primary dataset	Smartphone_Usage.csv – 7,500 records, 16 columns (CSV)